Accelerated II/III

Unit 0.5 HW#7 – Writing Equations of Piecewise Functions

Part I - Warm up: Match the piecewise function with its graph.

1. $f(x) = \begin{cases} x - 4, & \text{if } x \le 1 \\ 3x, & \text{if } x > 1 \end{cases}$

2. $f(x) = \begin{cases} 2x + 3, & \text{if } x \ge 0 \\ x + 4, & \text{if } x < 0 \end{cases}$

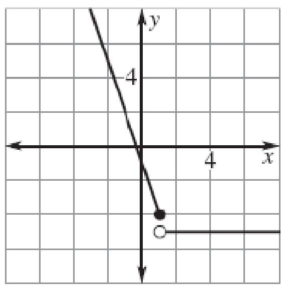
3. $f(x) = \begin{cases} x + 4, & \text{if } x \le 0 \\ 2x + 4, & \text{if } x > 0 \end{cases}$

4. $f(x) = \begin{cases} 3x - 1, & \text{if } x \ge -1 \\ -5, & \text{if } x < -1 \end{cases}$

5. $f(x) = \begin{cases} 3x - 2, & \text{if } x \le 1 \\ x + 2, & \text{if } x > 1 \end{cases}$

6. $f(x) = \begin{cases} -3x - 1, & \text{if } x \le 1 \\ -5, & \text{if } x > 1 \end{cases}$

A.



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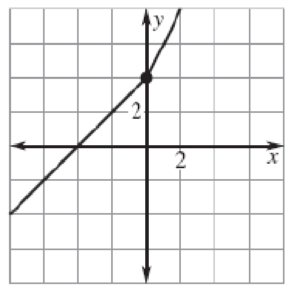

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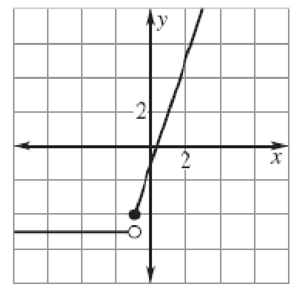

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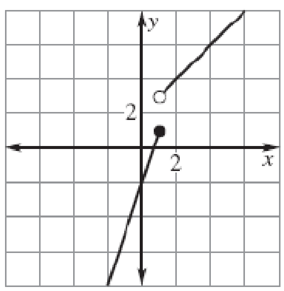

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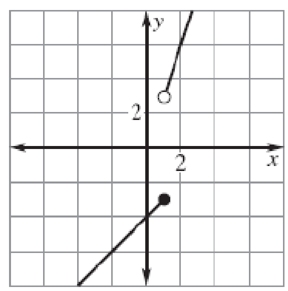
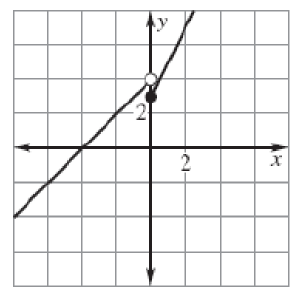
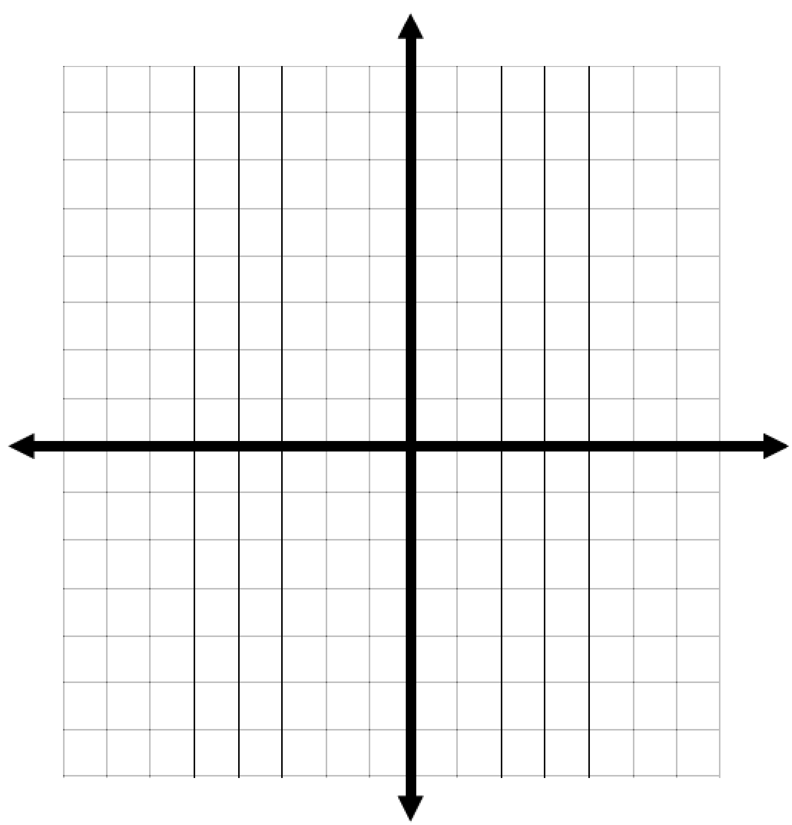


F.



Part II. Carefully graph each of the following. Identify whether or not the graph is a function. Then, evaluate the graph at any specified domain value.

7. $f(x) = \begin{cases} x + 5 & x < -2 \\ -2x - 1 & x \ge -1 \end{cases}$



Function? Yes or No

$f(3) =$

$f(-4) =$

$f(-2) =$

Domain:

Range:

8. $f(x)=\begin{cases} -x+4 & x\le 0 \\ \dfrac{2x}{3}-1 & 0<x\le 3 \\ 2 & x>5 \end{cases}$

Function? Yes or No

$f(-2)=$

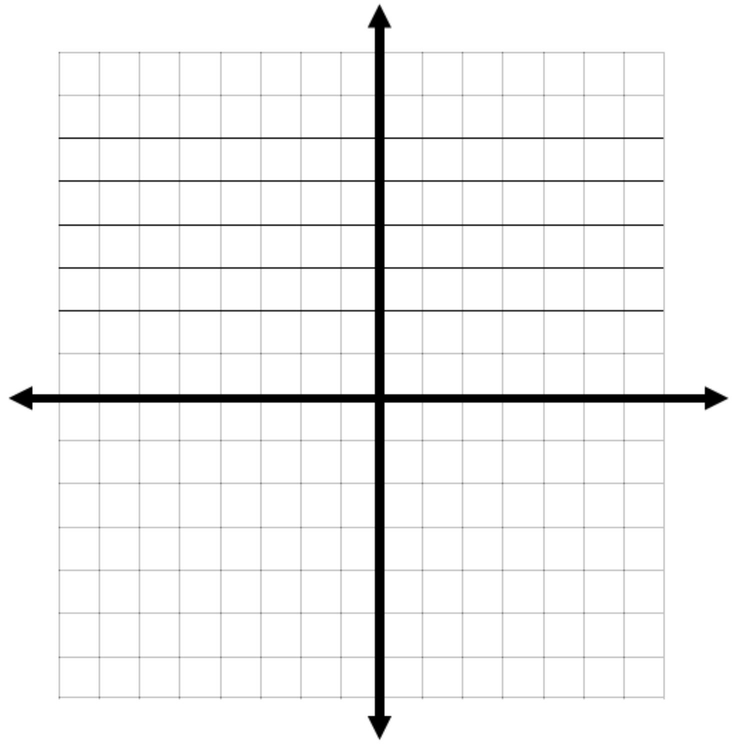
$f(0)=$

$f(5)=$

$f(7)=$

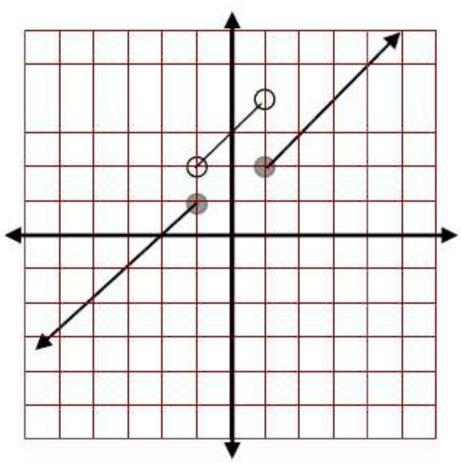
Domain:

Range:



Part III. Write equations for the piecewise functions whose graphs are shown below. Assume that the units are 1 for every square.

9.



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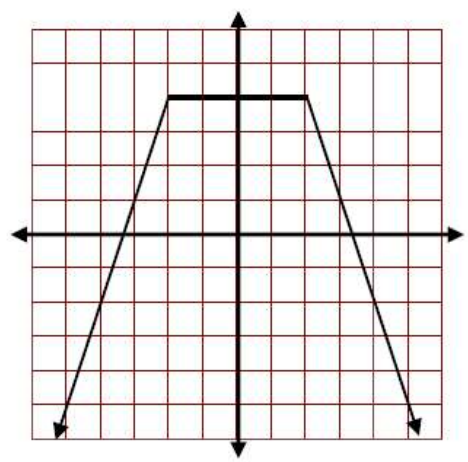

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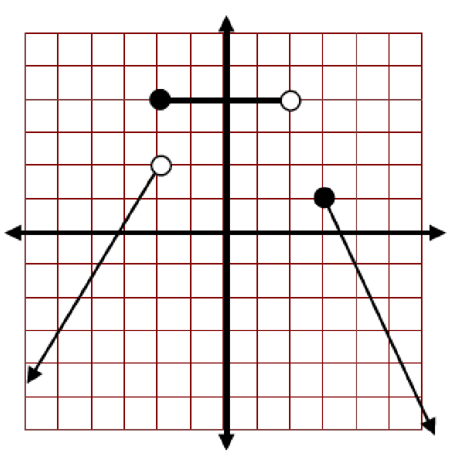

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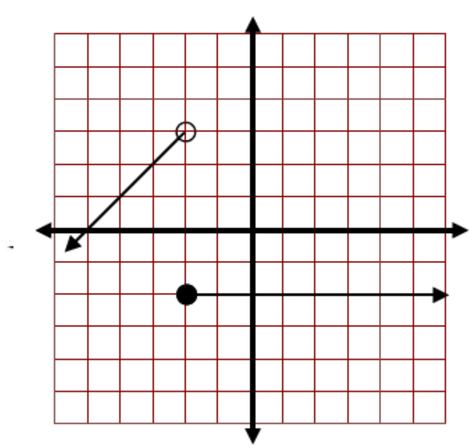


Answer key

Accelerated II/III

Unit 0.5 HW#7 - Writing Equations of Piecewise Functions

Part I - Warm up: Match the piecewise function with its graph.

E 1. $f(x) = \begin{cases} x - 4, \text{ if } x \le 1 \\ 3x, \text{ if } x > 1 \end{cases}$

F 2. $f(x) = \begin{cases} 2x + 3, \text{ if } x \ge 0 \\ x + 4, \text{ if } x < 0 \end{cases}$

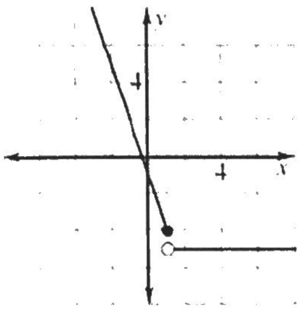
B 3. $f(x) = \begin{cases} x + 4, \text{ if } x < 0 \\ 2x + 4, \text{ if } x > 0 \end{cases}$

C 4. $f(x) = \begin{cases} 3x - 1, \text{ if } x \ge -1 \\ -5, \text{ if } x < -1 \end{cases}$

D 5. $f(x) = \begin{cases} 3x - 2, \text{ if } x < 1 \\ x + 2, \text{ if } x > 1 \end{cases}$

A 6. $f(x) = \begin{cases} -3x - 1, \text{ if } x \le 1 \\ -5, \text{ if } x > 1 \end{cases}$

A.



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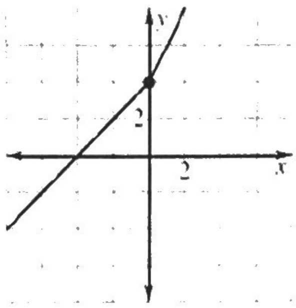

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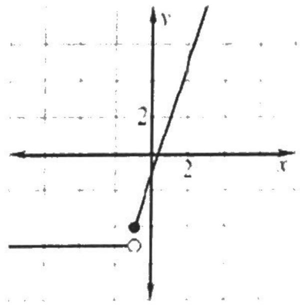

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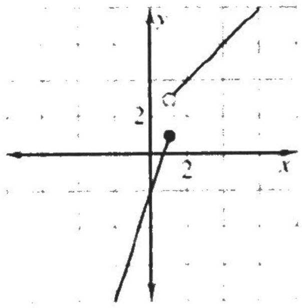

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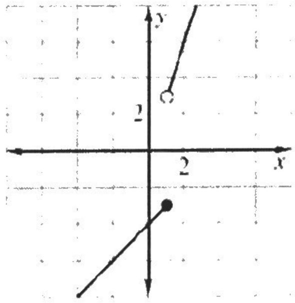

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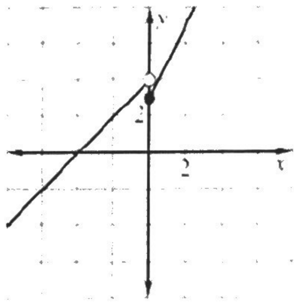
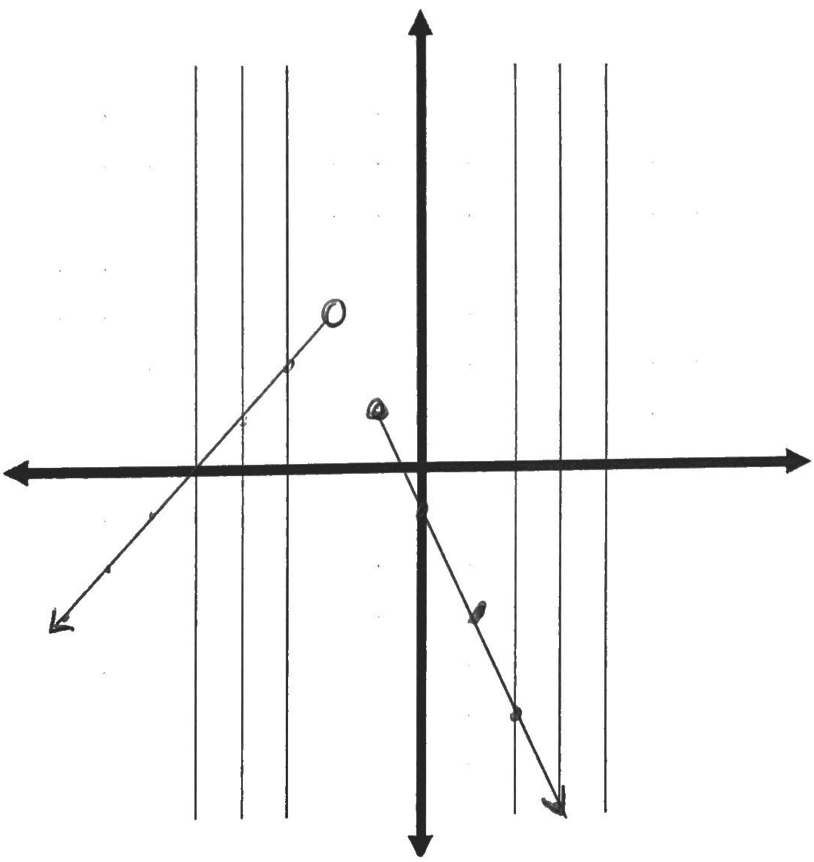


Part II. Carefully graph each of the following. Identify whether or not the graph is a function. Then, evaluate the graph at any specified domain value.

7. $f(x) = \begin{cases} x+5 & x < -2 \\ -2x-1 & x \ge -1 \end{cases}$



Function? Yes or No

$f(3) = -2(3) - 1 = -6 - 1 = \boxed{-7}$

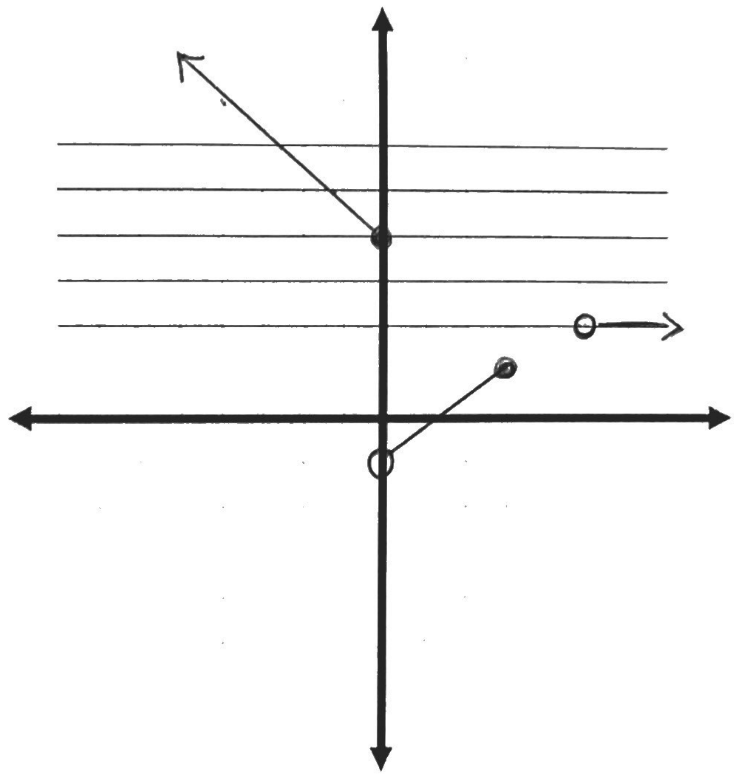
$f(-4) = -4 + 5 = \boxed{1}$

$f(-2) =$ undefined

Domain: $(-\infty, -2)$ $[-1, +\infty)$

Range: $(-\infty, 3)$

8. $f(x) = \begin{cases} -x+4 & x \le 0 \\ \frac{2x}{3} - 1 & 0 < x \le 3 \\ 2 & x > 5 \end{cases}$



Function? Yes or No

$f(-2) = -(-2) + 4 = 2+4 = \boxed{6}$

$f(0) = -(0) + 4 = \boxed{4}$

$f(5) = \boxed{\text{undefined}}$

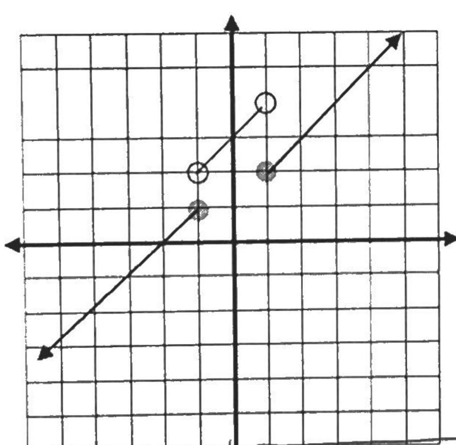
$f(7) = \boxed{2}$

Domain: $(-\infty, 3] ; (5, +\infty)$

Range: $(-1, 1]$ $[2]$ $[4, +\infty)$

Part III. Write equations for the piecewise functions whose graphs are shown below. Assume that the units are 1 for every square.

9.



$f(x) = \begin{cases} x+2; & x \le -1 \\ x+3; & -1 < x < 1 \\ x+1; & x \ge 1 \end{cases}$

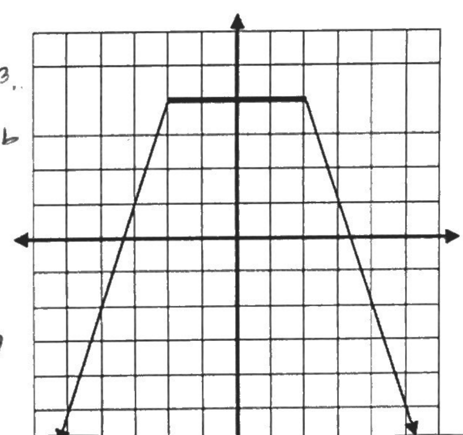
10.

$(-2, 4)$ $m = 3$

$4 = 3(-2) + b$

$4 = -6 + b$

$10 = b$



$f(x) = \begin{cases} 3x+10; & x \le -2 \\ 4; & -2 < x < 2 \\ -3x+10; & x \ge 2 \end{cases}$

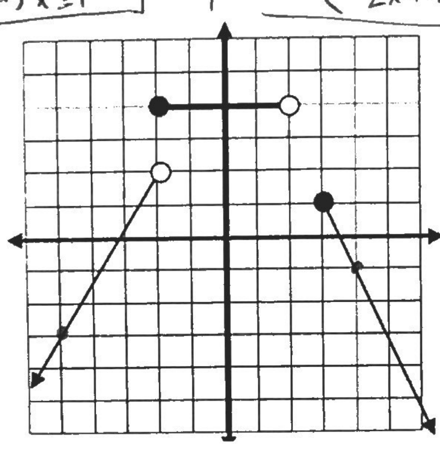
11.

$(-2, 2)$ $m = \frac{5}{3}$

$2 = \frac{5}{3}(-2) + b$

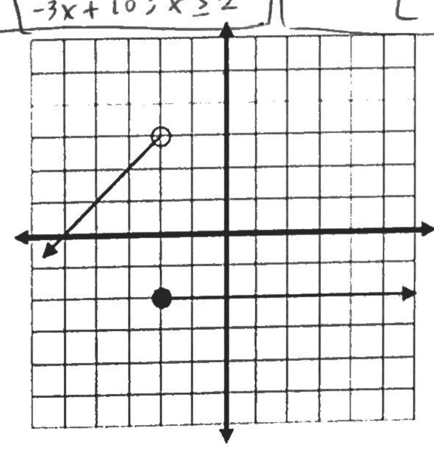
$\frac{6}{3} = -\frac{10}{3} + b$

$\frac{16}{3} = b$



$f(x) = \begin{cases} \frac{5}{3}x + \frac{16}{3}; & x < -2 \\ 4; & -2 \le x < 2 \\ -2x+7; & x \ge 3 \end{cases}$

12.



$f(x) = \begin{cases} x+5; & x < -2 \\ -2; & x \ge -2 \end{cases}$